Projective resolutions, surface orders and zeta determinants

Extended constructions, calculations and arithmetic research program

Research working draft

November 3, 2026

Abstract

This extended edition constructs forbidden-path projective resolutions and their graded Ext and Tor, proves the weighted Cartan determinant formula by finite arrow-state matrices, and develops the passage from derived objects to determinant lines and cohomological zeta functions. Examples isolate the effects of completion, singular seams, periodic tails, nonflat specialization and cancellation. We explain the graded-curve and Koszul-dual interfaces with their finiteness hypotheses, and compare finite-field purity with the missing global number-field positive determinant realization. Both successful algebraic constructions and unresolved arithmetic obligations are explicit. Selected components are verified in Lean4; neither the full derived framework nor GRH is formalized or proved.

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1 Objects, conventions and scope

The starting sources are Schreiber's surface-algebra and surface-order papers [1, 2], the generic-resolution discussion [3], and the weighted locally gentle determinant formula of Bessenrodt–Holm [4]. Our purpose is to supply the exact homological constructions needed to use their useful parts in a Hilbert–Pólya program. The source manuscripts are preserved; corrections below are statements in this new account. Full reading records, pinned versions and artifact hashes accompany the project. Related literature is used with the hypotheses stated here, rather than incorporated as an unqualified equivalence.

Fix a field k, a finite quiver $Q = (Q_0, Q_1, s, t)$, and an ideal I generated by a set of length-two paths. We compose paths in the order traveled: ab means first a, then b. Write $A = kQ/I$, e_i for its vertex idempotents, and $J = A_{>0}$ for the arrow ideal. A path is *permitted* when none of its adjacent pairs belongs to I. These paths, including the empty paths e_i, form a k-basis of A. The algebra may be infinite dimensional.

Definition 1.1. The pair (Q, I) is locally gentle if each vertex has at most two incoming and two outgoing arrows, and every arrow has at most one permitted successor, one forbidden successor, one permitted predecessor and one forbidden predecessor. A gentle algebra is additionally finite dimensional. We use right modules, $P_i = e_iA$ and $S_i = P_i/P_iJ$, with arrows of degree one and internal shift $(M\langle n\rangle)_d = M_{d-n}$.

Thus $P_i\langle n\rangle$ is generated in degree n; square brackets later denote the cohomological shift in a derived category. Completion $\widehat{A} = \varprojlim A/J^r$, a hereditary normalization of a completed order, and a central special fiber are different objects. A forbidden chain is a composable sequence all of whose adjacent pairs lie in I; it is generally zero as a path in A. We retain it as an indexing word. Forgetting this distinction makes the resolution below disappear.

The number-field target is initially primitive Dirichlet $L(s,\chi)$. Finite-order Hecke characters over a specified number field, Dedekind zeta functions, and general Artin representations require separate comparison theorems. General Artin holomorphy is not assumed. None of the algebraic results below identifies its determinant with any of these functions.

2 Minimal projective resolutions and the signed Euler matrix

For $n \geq 1$, let $\mathcal{R}_n(i)$ be the forbidden chains of length n starting at i; set $\mathcal{R}_0(i) = \{e_i\}$. For a nonempty chain $p = a_1 \cdots a_n$, write $p^- = a_1 \cdots a_{n-1}$.

Theorem 2.1 (Explicit linear resolution)**.** *For every quadratic monomial algebra as above, a graded projective resolution of S_i is*

$$\cdots \longrightarrow P_n(i) \xrightarrow{d_n} P_{n-1}(i) \longrightarrow \cdots \longrightarrow P_0(i) \longrightarrow S_i \longrightarrow 0, \qquad P_n(i) = \bigoplus_{p \in \mathcal{R}_n(i)} e_{t(p)} A\langle n \rangle.$$

On the p summand, $d_n(x) = a_n x$ in the p^- summand. The augmentation is the quotient $e_i A \to S_i$. The maps are right A-linear, homogeneous of degree zero, and valued in the graded arrow radical. The resolution is minimal in the graded sense. In particular A with its natural path grading is Koszul.

Proof. Each summand is an idempotent projective. The endpoints make left multiplication $a_n x$ a map into the indicated right module; associativity proves right linearity. The source generator has degree n, and its image has degree $1 + (n-1) = n$. Two successive maps multiply by $a_{n-1}a_n = 0$.

The augmentation kernel is the span of all nonempty permitted paths from i, the direct sum of aA over arrows a from i. This is $\operatorname{im} d_1$. More generally, the kernel of $e_{t(a)}A \xrightarrow{x \mapsto ax} e_{s(a)}A$ is the span of permitted paths whose first arrow b satisfies $ab \in I$. Indeed $ae_{t(a)} = a \neq 0$; every other basis path either vanishes after multiplication because of that first pair, or maps to a distinct surviving basis path. The kernel is therefore $\bigoplus_{b:ab\in I} bA$. For a fixed prefix p^-, images of distinct last arrows have disjoint nonzero path supports. There can be no cancellation between them. Taking their individual kernels identifies $\ker d_n$ with the image of the summands indexed by pb. This proves exactness in every degree without a global-dimension assumption. All entries have positive degree (graded minimality does not identify J with the ordinary Jacobson radical), so applying $\operatorname{Hom}_{\text{gr-}A}(-, S_j\langle r \rangle)$ kills the differentials. Minimality and linearity follow. The direct sum of the S_i is A/J, so its linear resolution is the Koszul property. □

Corollary 2.2 (Ext, Tor and inversion)*.* In ungraded modules, $\dim_k \operatorname{Ext}^n_A(S_i, S_j)$ is the number of forbidden chains of length n from i to j. The same number is $\dim_k \operatorname{Tor}^A_n(S_i, {}_jS)$, where ${}_jS = Ae_j/JAe_j$. With the above grading, Ext is supported on the matching internal shift n. Define

$$C_{ij}(z) = \sum_{d \geq 0} \dim_k (e_i A_d e_j) z^d, \qquad E_{ij}(z) = \sum_{n \geq 0} (-1)^n \dim_k \operatorname{Ext}^n_A(S_i, S_j) z^n.$$

Then $E(z)C(z) = C(z)E(z) = 1$ in $M_{Q_0}(\mathbb{Z}[[z]])$.

Proof. An A-map $e_v A \to S_j$ is determined by an element of $S_j e_v$, of dimension δ_{vj}. Each differential becomes zero since arrows annihilate the simple. Tensoring with a left simple gives the same calculation. Degree by degree, the augmented resolution has Euler characteristic S_i. Its nth term contributes $(-1)^n z^n C_{t(p),j}(z)$, so the alternating sum is exactly $(EC)_{ij} = \delta_{ij}$. Each coefficient is finite: $n \leq d$ and Q is finite. Since both matrices have constant term the identity, the formal matrix inverse is unique and two-sided. □

This already signed E is evaluated at z, not at $-z$. Other conventions can transpose the matrices, but cannot justify applying the sign twice. The complementary-relation algebra $A^{\#} = kQ/I^{\#}$ allows precisely the forbidden pairs of A, and $E(z) = C_{A^{\#}}(-z)$. Its paths encode the quadratic Koszul dual, with an opposite quiver if that convention is used for Yoneda composition. At the chain level Yoneda products are concatenation of forbidden chains (zero when the join is permitted), with the handedness/opposite convention just declared. An explicit comparison-map proof is given in the extended edition; the dimensions and inversion above do not require this product identification. Thus endpoint data and relations, rather than Betti numbers alone, control the product.

For locally gentle A, each initial arrow has at most one forbidden continuation. The resolution has at most two arms; they terminate or enter a finite cycle. A repeated ordered arrow pattern yields an eventual tail period after forgetting internal grading. This does not say that $\Omega^p S_i \cong S_i$, or that the displayed period is the least module-isomorphism period.

3 The weighted Cartan determinant from arrow states

Give each arrow an independent commuting weight w_a and work over $R = \mathbb{Z}[[w_a : a \in Q_1]]$. A path has weight the product of its arrows. The weighted Cartan matrix $C_{ij}(\mathbf{w})$ sums permitted paths from i to j. Finite Q guarantees finitely many terms of each total degree.

Put $n = |Q_0|$, $m = |Q_1|$ and define

$$S_{i,a} = \mathbf{1}_{s(a)=i}, \quad T_{a,j} = \mathbf{1}_{t(a)=j}, \quad D_{a,b} = \delta_{ab} w_a,$$

$$U_{a,b} = \mathbf{1}_{t(a)=s(b),\, ab \notin I} w_b, \qquad V_{a,b} = \mathbf{1}_{t(a)=s(b),\, ab \in I} w_b.$$

Here S is $n \times m$, T is $m \times n$, and U, V, D are $m \times m$. Every composable pair is exactly one kind, so $U + V = TSD$.

Theorem 3.1 (Weighted cycle formula, cf. Bessenrodt–Holm)**.** *For a finite locally gentle quiver,*

$$\det C_A(\mathbf{w}) = \frac{\displaystyle\prod_{c \in \mathcal{C}_{\mathrm{forb}}} (1 - (-1)^{\ell(c)} w(c))}{\displaystyle\prod_{c \in \mathcal{C}_{\mathrm{perm}}} (1 - w(c))}.$$

Both sets consist of primitive directed cycles of the corresponding arrow-transition graph, modulo cyclic rotation. Iterations are not extra factors. All denominators have constant term one.

Proof. Paths of length $r + 1$ contribute $SDU^r T$. Hence

$$C = 1 + SD(1 - U)^{-1} T, \qquad (1 - U)^{-1} = \sum_{r \geq 0} U^r.$$

The inverse converges in the augmentation topology because every entry of U has positive degree. Sylvester's determinant identity $\det(1 + AB) = \det(1 + BA)$ for rectangular A, B gives

$$\det C = \det\bigl(1 + (1 - U)^{-1} TSD\bigr) = \frac{\det(1 + V)}{\det(1 - U)}.$$

For completeness the rectangular identity follows by expanding the determinant of the block matrix $\left(\begin{smallmatrix} 1 & -A \\ B & 1 \end{smallmatrix}\right)$ with either identity block. Thus it is valid over any commutative ring. Multiplying before division gives a cleared identity even in more general coefficient rings.

Each vertex of the U transition graph has at most one predecessor and one successor, so every connected component is a directed chain or a directed cycle. The same holds for V. Ordering a chain along its arrows makes its transition matrix strictly triangular, contributing determinant one. On a cycle of length r, only the identity permutation and the full cyclic permutation contribute to $\det(1-U)$; their sum is $1-w(c)$. Replacing U by $-V$ gives $1-(-1)^r w(c)$. For a one-loop component this is simply $1-w_a$ or $1+w_a$, so there is no exceptional sign case. Multiplying the component determinants proves the formula, including quivers with both infinite permitted paths and infinite resolutions. □

The transfer identity before component factorization holds for every finite quadratic monomial quiver. The locally gentle hypothesis is used for the chain/cycle decomposition, not for Sylvester's identity. Formula 3.1 is a formal invariant, not the determinant of an infinite resolution. Specializing weights to a point where a denominator vanishes requires a separate cancellation or continuation argument.

4 Completion, singular seams and geometric objects

The node illustrates both the strength and the limits of the construction. For $A = k\langle x,y\rangle/(xy,yx) = k[x,y]/(xy)$,

$$C_A(w_x,w_y) = \frac{1-w_xw_y}{(1-w_x)(1-w_y)}, \quad C_A(z) = \frac{1+z}{1-z}, \quad E_A(z) = \frac{1-z}{1+z}.$$

The completed order $\Lambda = k[[x,y]]/(xy)$ has normalization $B = k[[x]] \times k[[y]]$ and conductor square

$$\Lambda = B \times_{k\times k} k,$$

where the lower map is diagonal and $B \to k \times k$ evaluates the two origins. Equality of constant terms proves this description: (f,g) corresponds uniquely to $f(x)+g(y)-f(0)$. No residue map $k((x)) \to k$ is involved; x is a unit in that field.

Its simple k has resolution

$$\cdots \xrightarrow{\mathrm{diag}(x,y)} \Lambda^2 \xrightarrow{\mathrm{diag}(y,x)} \Lambda^2 \xrightarrow{[x\ y]} \Lambda \longrightarrow k \longrightarrow 0.$$

Indeed $\mathrm{ann}(x) = (y)$, $\mathrm{ann}(y) = (x)$ and $(x) \cap (y) = 0$. With $t = x+y$, Λ is free of rank two over $k[[t]]$, whereas k is t-torsion. Positive syzygies are torsion-free over this DVR. Thus the simple is not periodic from degree zero. Moreover $(x) \oplus (y)$ and $(y) \oplus (x)$ are isomorphic: the unlabelled syzygy tail has period one even though the ordered branch pattern has period two. Tensoring with $k[[t]]/(t)$ does not preserve this resolution: the special fiber is $k[x]/(x^2)$, whose simple has Betti numbers $1,1,1,\ldots$, rather than $1,2,2,\ldots$. The missing homology is a Tor contribution.

For the following closed split subclass, require exactly two incoming and two outgoing arrows at each vertex, unique permitted and forbidden successors and predecessors, and two matched permitted-cycle incidences at each original vertex, with no unmatched endpoints. One can then specify hereditary cycle orders H_c over $k[[x_c]]$ and impose equality of the paired diagonal residues. Map each permitted cycle to successive matrix units, with its wrap arrow multiplied by x_c. Distinct permitted paths have independent coefficients in these entries, and every tuple with matching diagonal residues is reconstructed coefficient by coefficient. This proves the completed path-algebra/pullback comparison in this split setting. A common central parameter is $t \mapsto (x_c 1)_c$; long permitted paths show that the arrow-adic and t-adic filtrations are cofinal. Broader Dedekind bases need their own

residue identifications, local uniformizers and projective-cover hypotheses. Hereditary normalization does not make the glued order hereditary.

The derived category $D(A)$ is obtained from complexes of right A-modules by inverting quasi-isomorphisms. Its perfect subcategory $\mathrm{Perf}(A)$ consists of objects represented by bounded complexes of finitely generated projectives. A simple with an infinite minimal resolution need not be perfect. The homotopy category of bounded projectives embeds as $\mathrm{Perf}(A)$; $D^b(\text{mod-}A)$ can be larger. This is the correct finiteness test before forming finite determinants.

For finite-dimensional graded gentle algebras, the surface models of [6, 7, 8, 9] describe string objects by graded curves and band objects by closed curves with local systems. A line field/degree convention and endpoint or puncture behavior are essential. Appropriate oriented graded intersections give morphism bases, subject to exceptional closed-curve identity and Auslander–Reiten maps. Thus a bare count of transverse intersections is not a universal Hom formula. In a module heart, $\mathrm{Ext}^n(M,N) = \mathrm{Hom}_{D^b}(M, N[n])$; resolutions compute these groups and derived tensor products compute Tor. Rotating or zigzagging curves supplies concrete projective covers in the stated models. Koszul duality exchanges permitted and forbidden combinatorics through a derived functor with finiteness/completion conditions; a formal matrix inverse alone is not a derived equivalence. There is no justified transfer from these finite-dimensional categories to every completed arithmetic order, nor an identification of curve length with prime norm.

5 Determinant lines, chain endomorphisms and zeta functions

For a bounded complex $P^\bullet$ of finite projective modules over a commutative base R, its determinant is the graded invertible module

$$\det(P^\bullet) = \bigotimes_i (\det_R P^i)^{(-1)^i}.$$

It is a line, with the coherent signs of the determinant functor, not a scalar analytic function. Quasi-isomorphism invariance and derived base change hold in the perfect category [17, 18, 19]. For noncommutative A first choose a realization to a perfect complex over a commutative base; arbitrary A-projectives have no ordinary top-exterior-power determinant of this kind. A resolution is a computational presentation, not an additional arithmetic eigenvalue source.

Theorem 5.1 (Cohomological cancellation)**.** *Let $C^\bullet$ be a bounded finite-dimensional complex over a field and let F be a chain endomorphism. Then*

$$Z(C,F;u) := \prod_i \det(1 - uF \mid C^i)^{(-1)^{i+1}} = \prod_i \det(1 - uF \mid H^i(C))^{(-1)^{i+1}}.$$

An acyclic complex contributes one.

Proof. Set $B^i = \operatorname{im} d^{i-1}$ and $Z^i = \ker d^i$. Commutation with d preserves both. The two exact sequences $0 \to Z^i \to C^i \to B^{i+1} \to 0$ and $0 \to B^i \to Z^i \to H^i \to 0$ make F block triangular in adapted bases. Hence its characteristic determinant on C^i is the product of those on B^i, H^i, B^{i+1}. The two boundary factors have opposite adjacent exponents and cancel. The finite product requires no convergence argument. □

An infinite periodic resolution is not a bounded complex for this theorem. A truncation leaves a boundary syzygy; omitting it changes the object. Likewise the matrices U and $-V$ in Theorem 3.1 can package a formal virtual determinant, but have not been shown to be an arithmetic endomorphism commuting with the resolution differential. These are separate constructions.

Over a characteristic-zero field (or after extending scalars to a $\mathbb{Q}$-algebra), for a finite matrix M, differentiation of the determinant at the identity and the formal inverse series give

$$-\log\det(1-uM)=\sum_{r\geq 1}\frac{\operatorname{tr}(M^r)}{r}u^r.$$

Indeed the derivative of the left side is $\operatorname{tr}(M(1-uM)^{-1})$; expand and integrate term by term with constant term zero. For a partial permutation this becomes the primitive-cycle Euler product. Such transfer zeta functions are not automatically Hasse–Weil or number-field L-functions. An infinite-dimensional extension needs a stated Fredholm or regularized determinant and its convergence theorem.

5.1 Testing a positive Hodge replacement

A natural next attempt is to turn the resolution differential into a positive operator. The following calculation identifies exactly what this retains.

Theorem 5.2 (Finite Hodge determinant cancellation)**.** *Let $C^\bullet$ be a bounded finite-dimensional complex Hilbert complex, $D=d+d^*$ and $\Delta=D^2$. Write E,O for the even and odd sums, of dimensions e,o. Then*

$$\det(1+z\Delta_E)=\det(1+z\Delta_O),\qquad \mu^o\det(\mu 1_E+\Delta_E)=\mu^e\det(\mu 1_O+\Delta_O).$$

The shifted parity quotient is μ^{e-o} as a meromorphic function; pointwise division requires a nonzero denominator. Thus every nonzero spectral factor cancels in the plain alternating Hodge determinant.

Proof. In orthonormal bases the odd self-adjoint D has block form $\left(\begin{smallmatrix}0 & B^*\\ B & 0\end{smallmatrix}\right)$, so $\Delta_E=B^*B$ and $\Delta_O=BB^*$. Sylvester's identity applied to zB^*, B proves the first assertion. For $\mu\neq 0$, factor μ from each matrix and apply that identity at $z=\mu^{-1}$. The cleared equality is polynomial in μ, hence also holds at zero. The difference $e-o$ is the Euler characteristic, by cancellation of boundary dimensions. The quotient statement is meromorphic; it does not assign a pointwise value to a common $0/0$ at a negative Laplace eigenvalue. □

The ordinary determinant avoids this cancellation but is metric dependent even on the fixed exact complex $\mathbb{C}\xrightarrow{1}\mathbb{C}$: metrics $1, r>0$ give both Laplacians r, with ordinary factor $r-z$. Also $d(s)=s$ gives $\Delta(s)=|s|^2$, which is not holomorphic near $s=1$. Positivity of a Hodge construction therefore does not supply a determinant-preserving arithmetic comparison. Degree-weighted analytic torsion [24] is a different invariant and is not excluded by this unweighted parity calculation.

5.2 A relative boundary realization and its limit obstruction

Proposition 5.3 (Relative realization of the Cartan determinant)**.** *Let S be the unilateral shift on $\ell^2(\mathbb{N})$ and P_0 its initial rank-one projection. For a finite matrix M with $1-M$ invertible, define*

$$A_M=1-(S^*S)\otimes M,\qquad C_M=1-(SS^*)\otimes M.$$

Then $A_MC_M^{-1}=1-P_0\otimes M$, a finite-rank perturbation of the identity, and

$$R(M):=\det_F(A_MC_M^{-1})=\det(1-M).$$

Consequently $R(-V)/R(U)=\det C_A$ for the arrow-state matrices above, initially where the operators are defined and then by the indicated polynomial/meromorphic continuation.

Proof. Coordinates give $S^*S = 1$, $SS^* = 1 - P_0$. On $P_0 \otimes \mathbb{C}^r$, C_M is the identity and $A_M = 1 - M$; on its orthogonal complement both equal $1 \otimes (1 - M)$. Their relative product therefore has the stated form. Its Fredholm determinant is the finite determinant on its r-dimensional support. Theorem 3.1 gives the last assertion. Individual Fredholm determinants of the bulk operators are generally undefined; no quotient of two such determinants is taken. At singular $1 - M$ only the scalar polynomial continuation is asserted, not a defined quotient operator. □

This is an explicit Hilbert-space relative realization of the formal Cartan invariant, not an arithmetic or generally self-adjoint realization. Its natural finite sections expose a further analytic gap. For scalar $M = 1/2$, the truncated shift on $N \geq 2$ coordinates gives a relative operator equal to $1/2$ at the first coordinate, 2 at the last and 1 elsewhere. Its determinant is always 1. Extending it by the identity, it converges strongly to $1 - P_0/2$, whose Fredholm determinant is $1/2$. The difference from that limit is the last-coordinate projection, of trace norm exactly 1. Hence strong convergence here cannot justify determinant convergence. The extended edition gives the general matrix formula and the necessary infinite-dimensional qualifications.

6 What finite-field purity supplies

For a separated finite-type $X/\mathbb{F}_q$ and a suitable constructible ℓ-adic coefficient system $\mathcal{F}$, $\ell \neq \operatorname{char} \mathbb{F}_q$, the Grothendieck trace formula identifies

$$L(X, \mathcal{F}, u) = \prod_{x \text{ closed}} \det(1 - u^{\deg x} \operatorname{Fr}_x \mid \mathcal{F}_x)^{-1} = \prod_i \det(1 - u \operatorname{Fr} \mid H^i_c(X_{\overline{\mathbb{F}}_q}, \mathcal{F}))^{(-1)^{i+1}}.$$

We choose the Frobenius convention giving eigenvalue q on $H^2(\mathbb{P}^1)$. For a smooth projective geometrically connected curve and constant coefficients this is

$$Z(X, u) = \frac{\det(1 - u \operatorname{Fr} \mid H^1)}{(1 - u)(1 - qu)}.$$

Deligne's purity theorem gives every complex absolute value $\sqrt{q}$ on H^1 [20, 21]; with $u = q^{-s}$ the numerator zeros have $\operatorname{Re} s = 1/2$. This paragraph invokes deep external arithmetic theorems, not consequences of the elementary path-resolution proof. Weil II also treats morphisms over $\mathbb{Z}$ with fiberwise weights; its scope is not confined to one finite field. What is absent here is a global completed number-field determinant and a positivity theorem for its zero-bearing part. Open or singular spaces require mixed weights or intersection complexes: for example $Z(\mathbb{G}_m, u) = (1 - u)/(1 - qu)$ has a weight-zero zero.

One useful finite-field test is $X : y^2 = x^3 - x$ over $\mathbb{F}_5$. Its eight points give trace -2 and

$$Z(X, u) = \frac{1 + 2u + 5u^2}{(1 - u)(1 - 5u)}, \qquad \alpha_1, \alpha_2 = -1 \pm 2i,$$

with $|\alpha_i| = \sqrt{5}$. This benchmarks a genuine cohomological determinant. A nodal or gentle category is not automatically smooth and proper over k, the hypotheses of modern noncommutative Weil frameworks [22]. Surface topology alone supplies neither point counts nor this Frobenius realization.

Proposition 6.1 (Duality versus positive purity)**.** *An alternating duality $F^t J F = qJ$ does not force eigenvalues of F to have modulus $\sqrt{q}$. A positive Hermitian form H with $F^* H F = qH$, $q > 0$, does force that conclusion.*

Proof. For the first assertion take $q = 2$, $F = \mathrm{diag}(2, 1)$ and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Direct multiplication gives the duality, but its eigenvalues have squares of absolute values 4 and 1. For the second, if $Fv = \alpha v$ and $v \neq 0$, then $|\alpha|^2 v^* H v = (Fv)^* H (Fv) = q v^* H v$; divide by $v^* H v > 0$. □

Functional equations similarly pair zeros without placing them on a line. Local weight zero for finite-image Artin Frobenius, already known since its eigenvalues are roots of unity, is insufficient for global number-field GRH.

7 A precise arithmetic program and a closure theorem

Proposition 7.1 (Positive-length Cartan obstruction)**.** *In Theorem 3.1, specialize $w_a(s) = \eta_a e^{-s\lambda_a}$ with $|\eta_a| \leq 1$ and $\lambda_a > 0$. Its rational Cartan determinant and its reciprocal are holomorphic and nonzero on $\mathrm{Re}\, s > 0$. Neither, multiplied by holomorphic nonvanishing factors, can converge locally uniformly to a nonzero holomorphic target having a zero there.*

Proof. Every primitive cycle has positive total length and product weight of modulus strictly less than one in this half-plane. Each numerator and denominator factor is therefore nonzero; their finite product is holomorphic with holomorphic reciprocal. If a sequence of these functions converged locally uniformly to a nonzero target with a zero, choose a small closed disk around an isolated target zero and no boundary zero. Uniform boundary convergence and Rouché's theorem force an approximant zero in the disk, a contradiction. The identical argument handles the stated normalizations. □

Thus the resolution/Cartan formula rigorously diagnoses a failed direct Euler-product repair. Artificially centering the weights at $s - 1/2$ puts their elementary cycle zeros on a chosen line but proves no matching with an arithmetic L-function.

Theorem 7.2 (Conditional positive determinant closure)**.** *Let Ω be a connected open subset of $\mathbb{C}$ containing an open portion of the critical strip, and let $L_\circ$ be holomorphic on Ω, not identically zero. Suppose holomorphic functions D_N converge locally uniformly to $L_\circ$ and every zero of D_N in Ω lies on $\mathrm{Re}\, s = 1/2$. Then every zero of $L_\circ$ in Ω lies on that line. One sufficient finite realization is*

$$D_N(s) = g_N(s) \det\big(A_N - [-i(s - 1/2)]B_N\big), \quad A_N = A_N^*, \quad B_N > 0,$$

with g_N holomorphic and nonzero and the same convergence hypothesis.

Proof. At a singular pencil, a nonzero v satisfies $A_N v = z B_N v$. The quotient $z = (v^* A_N v)/(v^* B_N v)$ is real since its denominator is positive. For $z = -i(s - 1/2)$ this means $\mathrm{Re}\, s = 1/2$. If the limit had an off-line zero, a small disk avoiding the line and other zeros would contradict Rouché exactly as in Proposition 7.1. □

This theorem is a closure mechanism, not a construction of A_N, B_N, g_N. Our existing arithmetic partial-sum approximants converge with compact error bounds; no positive-pencil determinant identity for them has been proved. Replacing their matrices by Hermitian parts can change the determinant and is not such an identity.

A complete attempt using resolutions must discharge the following sequence of concrete claims.

1. Construct the arithmetic category and realization independently of unknown zeros. Specify coefficients, ramification, actual Galois/Frobenius action, archimedean data and conductor. A dessin's topological monodromy is not this action on one fixed algebra.

2. Choose a finite perfect realization, or an infinite analytic complex with a justified determinant. Prove chain-map compatibility and comparison through conductor gluing and derived base change; include boundary and singularity contributions. Retain determinant trivializations and normalization.

3. Prove the completed L-function identity, including gamma factors and multiplicities. Finite-dimensional finite-field rationality in u does not provide a finite-dimensional global Frobenius for primes of different norms. Number-field flow proposals [23] require a regularized determinant and a global generator.

4. Construct and prove positivity/polarization for the *same* zero-bearing realization, or prove a determinant-preserving positive-pencil comparison for the convergent arithmetic approximants. Alternating duality, local unitarity, positivity of unrelated moments and a formal Euler identity do not discharge this step.

5. Prove trace-norm or compact-uniform determinant convergence on an exhaustion, with a nonzero anchor and appropriate pole removal. Then apply Theorem 7.2, with each intended L-function scope and all exceptional points checked.

The local homological part can now be implemented explicitly. The global arithmetic identity, its positive realization and its analytic comparison remain mathematical research problems. No assumption of GRH, Artin holomorphy or unknown zero locations is used in the proved algebraic statements.

8 Verification and reproducibility

The separate pinned Lean4.24/mathlib layer verifies a general cleared version of the rectangular determinant identity, acyclic-pair characteristic-determinant cancellation, the actual alternating-duality counterexample, and exactness/projectivity components of a dual-number resolution over an actual nonsemisimple algebra. The Hodge module also verifies actual block squaring, Gram positivity and determinant cancellation, positive scalar metrics and their moving-zero example, and a universal ring identity for the boundary relative product. The infinite Hilbert-space/Fredholm and trace-norm arguments just given remain prose. Existing files verify positive Hermitian pencils and analytic nonvanishing components. Theorems print their types and foundation dependencies. These are components, not a formalization of all forbidden-chain resolutions, the derived surface models, the trace formula, Deligne purity, or GRH.

A standard-library exact rational checker tests finite quivers, path/chain coefficients, differentials and bounded internal-degree exactness, signed matrix inversion and the weighted cycle determinant, including nodal, terminating, periodic and cancellation examples. Finite tests supplement the universal proofs; they do not certify the arithmetic bridge. All source submodules and pinned environments are preserved. Both editions compile from shared core text; the extended edition adds explicit computations, geometric hypotheses and implementation tasks.

A Worked resolutions and matrices

A.1 A terminating two-arrow example

Take $1 \xrightarrow{a} 2 \xrightarrow{b} 3$ with $ab = 0$. Then

$$0 \longrightarrow P_3\langle 2\rangle \xrightarrow{x\mapsto bx} P_2\langle 1\rangle \xrightarrow{x\mapsto ax} P_1 \longrightarrow S_1 \longrightarrow 0.$$

The permitted path bases are $P_1 = \langle e_1, a\rangle$, $P_2 = \langle e_2, b\rangle$, $P_3 = \langle e_3\rangle$. The second map has kernel kb, the image of the first, and the first is injective. Consequently

$$C(z) = \begin{pmatrix} 1 & z & 0 \\ 0 & 1 & z \\ 0 & 0 & 1 \end{pmatrix}, \qquad E(z) = \begin{pmatrix} 1 & -z & z^2 \\ 0 & 1 & -z \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiplication gives the identity: the $(1,3)$ coefficient is $-z^2+z^2 = 0$. In particular $\mathrm{Ext}^2(S_1, S_3) = k$, although $\mathrm{Hom}(S_1, S_3) = 0$. Its generator is the Yoneda splice of the two arrow extensions. In the complementary algebra the path ab survives; this is the simplest nontrivial product in the quadratic dual. All determinant factors here are one since neither transition graph has a cycle.

A.2 Comparison maps and completed resolutions

Here is the promised Yoneda calculation with bases fixed. For a forbidden chain p of length m from i to j, let $f_p : P_m(i) \to S_j$ project to the summand indexed by p and apply its simple augmentation. Lift it for every $r \geq 0$ to

$$F_p^r : P_{m+r}(i) \longrightarrow P_r(j)$$

by sending the summand indexed by pq identically to the summand indexed by q, and every other summand to zero. The zero-length suffix has endpoint j. A summand pq exists only if the entire concatenation is a forbidden chain. For $r \geq 1$, removing its last arrow commutes with removing the fixed prefix p; both composites multiply by that same last arrow. Summands with a different prefix remain zero on both sides. Thus these are comparison maps of the homological tails, lifting f_p through the augmentation. Composing F_p^n with $f_q : P_n(j) \to S_l$ picks exactly the summand pq in degree $m+n$. If the joining pair is permitted that summand does not exist, so the composite is zero. These representatives compute the Yoneda splice. With the corresponding homological convention their nonzero coefficient is one; converting to signed cochain shifts transports the usual shift signs consistently. This proves the stated multiplication rule, not merely its dimensions.

The path argument also extends to the arrow-adic completion, but should be proved directly rather than justified by an unproved flatness assertion. An element of $e_v\widehat{A}$ is a formal sum of permitted paths, with finitely many paths in every bounded length. Multiplication by an arrow either kills a coefficient because its first pair is forbidden or prefixes a distinct permitted path. Its kernel is exactly the finite sum of the formal branch ideals $b\widehat{A}$ for forbidden successors b. Different branches have disjoint supports, and every formal series on such a branch is obtained by factoring off its first arrow. Each map and its image is closed in the length topology. Consequently the completed version of Theorem 2.1, using finite direct sums at each fixed homological degree, is exact coefficient by coefficient. This is a filtered resolution; its associated grading has the internal shifts used above.

The closed arrow ideal $\widehat{J}$ is the Jacobson radical: for $x \in \widehat{J}$ and any $a \in \widehat{A}$, the geometric series in ax inverts $1 - ax$, proving $\widehat{J} \subseteq \mathrm{Jac}(\widehat{A})$, and the quotient is the semisimple algebra k^{Q_0}, proving the reverse inclusion. The augmentation maps are therefore projective covers and the completed resolution is minimal in this semiperfect setting. Applying Hom to vertex simples again kills every differential. These conclusions concern the specified completion over k, not arbitrary changes of coefficient rings or central specialization.

A.3 Dual numbers, actual exactness and nonperfectness

For $R = k[\varepsilon]/(\varepsilon^2)$ the maps in

$$\cdots \xrightarrow{\varepsilon} R \xrightarrow{\varepsilon} R \xrightarrow{\varepsilon} R \xrightarrow{\pi} k \longrightarrow 0$$

are R-linear. Write $r = a + b\varepsilon$. Then $\varepsilon r = a\varepsilon$, so $\ker(\varepsilon\cdot) = k\varepsilon = \operatorname{im}(\varepsilon\cdot)$ and $\ker \pi = k\varepsilon$. The preimage of $b\varepsilon$ is the scalar b; the preimage of $a \in k$ under π is $a \in R$. This proves exactness in all degrees and augmentation surjectivity. Every term is the regular free rank-one R-module. Applying $\operatorname{Hom}_R(-, k)$ makes every arrow zero and each term is k, since an R-linear map from R is determined freely by its value at one. Thus

$$\operatorname{Ext}^n_R(k,k) = k, \quad \operatorname{Tor}^R_n(k,k) = k \quad (n \geq 0), \qquad C_R(z) = 1 + z, \quad E_R(z) = \frac{1}{1+z}.$$

Nonvanishing Ext in all degrees rules out a finite projective resolution of k, so $k \notin \operatorname{Perf}(R)$. Infinite periodicity cannot be used to replace it by an arbitrary finite perfect complex. In this example the permitted transition is zero and the forbidden transition is the one-loop ε; the determinant formula gives $1 + w_\varepsilon$ directly.

The Lean implementation uses the actual algebra `QuadraticAlgebra k 0 0`, not a coordinate pair with an assumed differential. It selects the regular self-action explicitly and the residue action pulled back along the augmentation. The proof of all-degree exactness is the coordinate calculation just made, with an explicit preimage. It also proves each term projective and every induced Hom boundary zero, but does not yet identify the abstract derived Ext functor or construct mathlib's categorical resolution object. Those distinctions are recorded with the theorem types.

A.4 The node and its complementary relations

In $A = k[x,y]/(xy)$ the nonempty permitted paths are x^r, y^r. The forbidden chains alternate x, y, so there are two in every positive length. Summation gives

$$C_A(z) = 1 + 2\sum_{r\geq 1} z^r = \frac{1+z}{1-z}, \quad E_A(z) = 1 + 2\sum_{r\geq 1}(-z)^r = \frac{1-z}{1+z}.$$

The unsigned Ext series is again $(1+z)/(1-z)$. The forbidden cycle has length two and weight $w_x w_y$, while the two permitted loops have weights w_x, w_y, producing the multivariate formula in the core. Applying an extra sign to the already signed Euler series gives

$$C_A(z)E_A(-z) = \left(\frac{1+z}{1-z}\right)^2 = 1 + 4z + 8z^2 + \cdots \neq 1.$$

This detects the original preprint's sign error independently of any matrix transpose.

The complementary algebra $A^\# = k\langle x,y\rangle/(x^2, y^2)$ has two forbidden loops and one permitted alternating cycle, hence

$$C_{A^\#}(w_x, w_y) = \frac{(1+w_x)(1+w_y)}{1 - w_x w_y}.$$

After $w_x = w_y = z$ it has the same univariate Hilbert and Betti dimensions as the node, while its maps and relations differ. The equality of these coarse numerical series does not determine the algebra or the surface gluing.

For the completed node, let $R_0 = k[[t]]$ and $t = x + y$. Then

$$\Lambda \cong R_0[x]/(x^2 - tx) = R_0 \oplus R_0 x, \qquad y = t - x.$$

The displayed basis follows by reducing powers of x and comparing the two branch series. Each positive syzygy, a submodule of a finite free R_0-module, is R_0-torsion-free. Since t kills the simple, no positive syzygy is isomorphic to that simple. The labelled branch maps alternate, but the direct

sum of the two syzygy branches can be interchanged. Thus a labelled return time and a least ungraded module period are different invariants. In the graded category the internal shifts must be retained as well.

The branch module $M = \Lambda/(y)$ has alternating resolution

$$\cdots \xrightarrow{y} \Lambda \xrightarrow{x} \Lambda \xrightarrow{y} \Lambda \longrightarrow M \longrightarrow 0.$$

Its exactness is the same annihilator computation. Tensoring with k gives a zero differential in every degree, so $\operatorname{Tor}_n^{\Lambda}(M,k) = k$ for all n. It is not perfect, although the normalization factors $k[[x]]$ and $k[[y]]$ are regular hereditary rings. This is the elementary matrix-factorization mechanism over the hypersurface xy [14].

A.5 A nontrivial cancellation invisible to the determinant

Let $a, b : 0 \to 1$ and $c, d : 1 \to 0$. Declare the permitted transitions to be

$$a \to c \to b \to d \to a$$

and the forbidden transitions to be

$$a \to d \to b \to c \to a.$$

Both cycles have length four and the same commuting product $w_a w_b w_c w_d$. The numerator and denominator of Theorem 3.1 cancel, so $\det C = 1$ as a formal series. With all weights z one can see the entire matrix:

$$C(z) = \frac{1}{1-z^2}\begin{pmatrix} 1+z^2 & 2z \\ 2z & 1+z^2 \end{pmatrix}, \quad E(z) = \frac{1}{1-z^2}\begin{pmatrix} 1+z^2 & -2z \\ -2z & 1+z^2 \end{pmatrix}.$$

Their product is the identity, their determinants are one, and both simples have two resolution arms forever. In particular a scalar determinant neither detects finite projective dimension nor recovers the endpoint-refined Ext algebra. Any arithmetic comparison must specify which information survives its realization.

B Exactness in parameter families and correct specialization

B.1 What a generic complex does and does not prove

A family of matrices satisfying $d_i d_{i+1} = 0$ is a family of complexes. Adding rank upper bounds does not imply exactness: zero matrices satisfy all such equations. For a finite free complex over a commutative Noetherian ring B, the Buchsbaum–Eisenbud criterion [12] uses expected ranks

$$r_i = \sum_{j=i}^{n} (-1)^{j-i} \operatorname{rank} F_j$$

and the conditions $\operatorname{rank} d_i = r_i$ and $\operatorname{grade} I_{r_i}(d_i) \geq i$ for $i \geq 1$, with the usual zero-rank conventions. There is no requirement that d_1 surject onto F_0: its cokernel is the resolved module. Over a general Noetherian ring height is not a substitute for grade. For constant-rank finite projectives one works locally. This is an imported commutative acyclicity theorem, not a theorem about minors of arbitrary noncommutative arrow entries.

Theorem 5.3 of [3] does not state this criterion. Its discussion of generic resolutions must also be read with Bruns's direct-limit construction [13], rather than a claim that an unspecified finite exactification stage always terminates. A generic based resolution specializes to prescribed resolutions under the construction's hypotheses; arbitrary points of a rank-bounded complex scheme need not be such resolutions.

For a gentle algebra over k, use actual idempotent projectives and their path maps. If a parameter algebra A_B is finite projective over a commutative base B, the underlying B-complex of finite A_B-projectives is finite projective. One may apply the commutative criterion to that realization, or prove flatness and Tor vanishing. For infinite-dimensional completions neither finite rank nor finite length is automatic.

B.2 A complete base-change calculation

Over $B = k[t]$, the complex $B \xrightarrow{t} B$ in degrees $-1, 0$ resolves $k = B/(t)$. Tensor with k: the map becomes zero, giving $H^{-1} = k$ as well as $H^0 = k$. The two-term complex is the correct *derived* base change; replacing it by k in degree zero loses Tor. This example proves why arbitrary specialization is invalid without appealing to a complicated generic ring.

For the node, set $t = 0$ in the R_0-free order resolution. Its first maps become

$$\overline{\Lambda}^2 \xrightarrow{\operatorname{diag}(-x,x)} \overline{\Lambda}^2 \xrightarrow{[x\ -x]} \overline{\Lambda}, \qquad \overline{\Lambda} = k[x]/(x^2).$$

Write a vector as $(a + bx, c + dx)$. Its image under the row map is $(a - c)x$, so the kernel has dimension three over k, while the diagonal image has dimension two. Thus H_1 has dimension one. The augmentation cokernel has dimension one and the higher positive homology vanishes, as also follows from $\operatorname{Tor}_*^{R_0}(k, k)$ using the two-term R_0 resolution. The rank-one dual-number resolution of the special-fiber simple is a different exact complex.

The same care applies to hereditary orders. If $H = \left(\begin{smallmatrix} R_0 & tR_0 \\ R_0 & R_0 \end{smallmatrix}\right)$, then H/tH has dimension four, not three: the upper-right basis vector tE_{12} survives modulo tH. Entrywise reduction into $M_2(k)$ kills that vector and gives a different triangular image. A special fiber is the quotient of the module by its central parameter, not an informal entrywise picture.

B.3 Rank loci and a corrected determinantal example

The rank-at-most-r locus is cut out by $(r + 1)$-minors, not r-minors. In the one-arrow format of dimensions $(1, 1)$ and upper rank $r = 1$, the coordinate ring is simply $k[x]$: there are no 2-minors. Excluding the size-one maximal minor x would leave only the monomial 1, which is not a basis for $k[x]$. This is a decisive test of Definition 8.5 in [3]. More generally equality of complementary minors requires an actual complex, expected-rank/exactness hypotheses, determinant-line orientations and multipliers. For independent matrices $A = [I_3\ 0]$ of size 3×5 and $B = 0$ of size 5×2, $AB = 0$, a 3-minor of A is one and every 2-minor of B is zero. Their exterior-power matrices cannot be unconditionally equal.

A unit in a differential of a free complex, on the other hand, is not by itself a mistake. Elementary basis changes split off an identity map $B \xrightarrow{1} B$ in adjacent degrees; this pair is contractible and can be canceled. The nonminimal format $(1, 6, 8, 4, 1)$ in [3] reduces in this way to $(1, 6, 8, 3)$ in its determinantal example. Theorem 5.1 explains why a compatible chain endomorphism on that acyclic pair cannot create a new zeta factor.

C Derived geometry, curve computations and Koszul interfaces

C.1 From complexes to a usable category

A cochain complex has maps $d^i : C^i \to C^{i+1}$ with $d^{i+1}d^i = 0$. Its homotopy category identifies maps differing by $dh + hd$. A quasi-isomorphism induces isomorphisms on every cohomology group. Localizing at these maps gives $D(A)$. A bounded-above projective resolution computes $\mathbf{R}\operatorname{Hom}$ from a module, while a bounded perfect replacement permits finite determinant operations. These are different finiteness demands.

For a right module M and a left module N, choose the former's projective resolution $P_\bullet$; $H_n(P_\bullet \otimes_A N)$ is $\operatorname{Tor}_n^A(M, N)$. For right M, N, $H^n \operatorname{Hom}_A(P_\bullet, N)$ is $\operatorname{Ext}_A^n(M, N)$. Theorem 2.1 makes both computations explicit for vertex simples. An arbitrary string module requires the actual overlap/projective-cover maps, not just its intersection count.

Under our natural grading, a relation word and an allowed path in $A^\#$ are the same indexing sequence. Choosing dual arrow functionals yields the quadratic dual with reversed arrow orientation in the usual tensor-dual convention. This accounts for apparent opposites in Ext-algebra formulas. The functorial Koszul construction uses a graded/projective resolution of A/J and its dg endomorphisms; passing from this dg algebra to its cohomology and asserting an equivalence requires the proper Koszul and finiteness/completion hypotheses. It does not follow from equality of Hilbert series: the node and its complementary algebra already warn against reconstructing a category from that series.

C.2 A practical curve-to-resolution workflow

In the finite-dimensional graded gentle setting, a marked surface with dissection and line field encodes the algebra. Crossing the dissection records the combinatorial word and its degrees. For module curves in a specified heart this is an ordinary string; general derived objects use the appropriate homotopy-string or homotopy-band data, not necessarily a module string. At each crossing the line field fixes the grading. Boundary endpoints and punctures govern finite versus infinite tails. Closed curves additionally carry a local system and can represent band objects. A topological curve is used up to the appropriate homotopy, often drawn in minimal position. A choice of hyperbolic metric can give a geodesic representative but is not the definition of its algebraic degree or an arithmetic prime length.

An ordinary string is a composable word in arrows and formal inverse letters, with no immediate cancellation and no subword belonging to I or the inverse of such a subword. Its module has one basis vector for each position of the word, placed at the corresponding vertex; each traversed arrow carries the specified adjacent vector to the next (or previous for an inverse traversal), and all other entries are zero. Repeated vertices give distinct basis vectors, not an identification of positions. A band is an appropriate primitive cyclic string whose powers remain strings; a local system supplies its cyclic gluing and parameter. Formal inverse letters record a linear map's direction in the word, not invertible elements of A. Derived homotopy-string conventions allow different relation behavior and must be taken from the chosen model. This is why a general derived curve is not automatically a module in degree zero.

The following computation is supported by the precise models, rather than a proposed new equivalence.

1. Fix the algebra, its handedness, the marked surface, line field and the exact object category in [6]. Record whether the curve gives a perfect complex or an object with an infinite tail.

2. Use the zigzag/projective-curve construction of Chang [7], Theorem 2.16 and Proposition 2.23, or the rotated construction of [8], to list idempotent-projective summands with degrees and actual maps. At a puncture test the infinite periodic-tail case before applying a finite determinant. Chang's Remark 2.27 and Corollary 2.28 distinguish those possibilities.

3. Put curves in the prescribed minimal position and retain oriented, graded intersections. Under [6], Theorem 3.3, these label the standard morphisms, with extra identity and Auslander–Reiten connecting maps in the exceptional coincident closed-curve cases. Local systems modify the closed-curve calculation. Ordinary ungraded transverse crossings omit these data.

4. Compare these maps to $\mathrm{Hom}(M, N[n])$ in the chosen heart and hence to Ext. For Tor specify a left module or dual realization first; intersections between two right-module curves are not already a tensor product.

5. If using geometric Koszul duality, verify the homologically smooth graded gentle hypotheses of [9], whose body Theorems 3.6 and3.9 implement the half-rotation functor and indexed intersection formula. Do not import that equivalence to a singular completed order without a comparison theorem.

For the terminating example above these operations must recover three projective curves, the maps a, b, a nonzero degree-two morphism from S_1 to S_3, and zero ordinary Hom. This provides a small acceptance test with known exact answer. The project currently proves the algebraic calculation; a full curve-diagram implementation is a next construction, not an artifact silently claimed complete.

Surface cuts and recollements [10] can organize exact triangles, whose realized determinant lines multiply. Their hypotheses on the generated subsurface and interior marked points must be retained. The recent locally gentle resolutions [5] give a direct independent algebraic reference; recent gentle-tree resolving subcategories [11] organize syzygy/extension closures but do not make every closed surface order smooth. These are complementary tools for selecting a perfect subcategory and checking which arithmetic operations preserve it.

C.3 Patching and the center

Projective modules over a conductor fiber product can be patched from projectives on its two pieces and an actual isomorphism on the overlap, under the projective-patching hypotheses. Beauville–Laszlo [15] is a precise alternative for vector bundles with a non-zero-divisor parameter and specified localization/completion data. Its descent diagram is part of the theorem; equality of a few diagonal entries is not a proof of a global affine-Schubert bundle or Langlands equivalence.

Likewise for GL_n over $F = k((t))$, $R_0 = k[[t]]$, distinguish

$$\mathrm{Gr} = \mathrm{GL}_n(F)/\mathrm{GL}_n(R_0), \qquad \mathrm{Fl}_{\mathrm{aff}} = \mathrm{GL}_n(F)/\mathcal{I}, \qquad \mathrm{GL}_n(R_0)/\mathcal{I} = \mathrm{GL}_n(k)/B.$$

Here $\mathcal{I}$ is the Iwahori *group*, not the entire order. Full lattice chains consist of rank-n lattices with successive residue dimensions one. Nilpotent cyclic-quiver representations can be realized by the appropriate resolvent/lattice map into Schubert geometry [16]; nilpotency of cyclic products is weaker than requiring those products to vanish. This controls local algebraic strata, not arithmetic Frobenius or the compatibility of every seam in a proposed global construction.

Here is the coefficientwise proof of the completed-order comparison used in the core. For a permitted cycle of length n_c, choose positions $1, \dots, n_c$ and let $H_c \subset M_{n_c}(k[[x_c]])$ be generated by the diagonal idempotents, $E_{j,j+1}$ for $j < n_c$, and $x_c E_{n_c,1}$. In this convention its upper-triangular entries lie in $k[[x_c]]$ and its strictly lower entries in $x_c k[[x_c]]$. Reversing the indexing gives the other

usual Iwahori-order convention. The radical consists of all off-diagonal entries together with diagonal entries divisible by x_c, and $H_c/\operatorname{rad} H_c = k^{n_c}$.

The closed incidence hypothesis pairs exactly two cycle positions over each original quiver vertex, possibly on the same cycle. Form the subalgebra Λ' of $\prod_c H_c$ where their diagonal constants agree. Send e_v to the sum of its two position idempotents and send each arrow to its successive matrix unit, with the wrap multiplier x_c. Forbidden pairs vanish because they change permitted cycle positions without a matching matrix index. For every fixed pair of positions, the permitted paths are the unique oriented segments followed by any number of full turns. Their images are exactly the corresponding distinct powers of x_c times the matrix unit. A positive path returning to its initial position has at least one full turn, so its diagonal entry has zero constant term. Thus no positive path can cancel a vertex constant or another path coefficient. This proves injectivity on formal series. Conversely any tuple in Λ' decomposes into its matching vertex constants and its independent positive path series in these entries. This proves surjectivity and establishes the actual algebra isomorphism $\widehat{A} \cong \Lambda'$.

Put $L = \max_c n_c$. The element $t = (x_c 1)_c$ belongs to the image and is central. Any permitted path of length at least $L+1$ makes a full turn on its chosen cycle and leaves a nonempty suffix. Multiplying that suffix by t adds exactly the full turn on its branch: the other cycle incidence at its initial vertex is annihilated by the forbidden relation. Factoring coefficients on each branch therefore gives $\widehat{J}^{L+1} \subseteq t\Lambda'$, whereas $t\Lambda' \subseteq \widehat{J}$. Raising these inclusions proves cofinality of the two filtrations. Each H_c is finite free over the common $k[[t]]$ with $x_c = t$; Λ' contains $t\prod_c H_c$ and differs only by finitely many residue equalities. It is finite and torsion-free over this DVR, hence free, of rank $\sum_c n_c^2$. This proves the split common-base assertion with no assumption that the cycle lengths agree. It does not establish a mixed-characteristic or arbitrary Dedekind-base comparison.

A further useful repair concerns central data. In a connected quiver, $\sum c_i e_i$ commutes with each arrow only if its coefficients agree along that arrow, hence all c_i are equal. Thus the entire degree-zero algebra kQ_0 is not central. In the split closed case, sums of all rotations of each permitted cycle give the positive central generators; one must sum rotations, since an individual based cycle need not commute with arrows. More importantly normalization remembers permitted-cycle lengths but not all gluing. For example put $\sigma = (123)(456)$. The pairings

$$\alpha_1 = (14)(25)(36), \qquad \alpha_2 = (14)(26)(35)$$

give $\phi_j = \alpha_j \sigma^{-1}$ with respectively one six-cycle and three two-cycles. Both connected dessins have normalization type $H_3 \times H_3$ but genera one and zero. Indeed $V = 2$, $E = 3$ and $F = 1$ or 3, so $2 - 2g = V - E + F$. A normalization invariant cannot determine the passport, much less a Galois orbit. The endpoint-refined resolution maps and actual incidence are necessary.

D Cartan–Eilenberg, Weil models and Frobenius: three roles

The phrase Cartan–Weil can refer to different useful structures; keeping their roles distinct prevents an erroneous proof transfer. Cartan–Eilenberg homological algebra controls double resolutions, derived functors and spectral sequences. A first-quadrant double complex has a convergent filtration spectral sequence because each total degree has finitely many entries; unbounded totalizations require a separate convergence/completion choice. This matters for periodic surface-order resolutions. Finite derived constructions do not justify exchanging arbitrary infinite sums, determinants and specializations.

The Cartan/Weil models of equivariant de Rham theory encode connections, curvature and

characteristic classes. In the one-dimensional abelian Weil algebra over characteristic zero,

$$W = k[u] \otimes \Lambda(\theta), \qquad |\theta| = 1, \quad |u| = 2, \quad d\theta = u, \quad du = 0.$$

Every element is $f(u) + \theta g(u)$ and its differential is $ug(u)$. Since u is not a zero divisor, the closed elements are $k[u]$ and the boundaries are $uk[u]$, so $H(W) = k$. The basic subcomplex in equivariant theory is a separate construction and need not have the same cohomology. This is a legitimate acyclic-resolution model, unlike taking $d\varepsilon = 1$ and claiming cohomology k.

Indeed in the source DG example $k[\varepsilon]/(\varepsilon^2)$ with $|\varepsilon| = -1$ and $d\varepsilon = 1$, a closed z satisfies $d(\varepsilon z) = z$. Its cohomology is zero and no augmentation to k can be a DG map since the unit is a boundary. This is compatible with $d^2 = 0$; square-zero alone does not identify the intended resolved object. Our dual-number resolution uses an ordinary algebra with zero algebra differential and a separate module differential given by multiplication by ε, so it does not repeat that mistake.

Weil cohomology and ℓ-adic Frobenius play a third role. They supply arithmetic traces and weights, not merely a resolution of the unit. Grothendieck's trace formula and Deligne's weight theorem are the two deep inputs in the finite-field calculation in the core. Weil II Theorem 3.3.1 is formulated for finite-type morphisms over $\mathbb{Z}$ with fiberwise weights; Corollaries 3.3.4–3.3.6 control compact-support/ordinary cohomology and purity of the image $H^i_c \to H^i$. This is not a global complex-variable determinant formula for completed number-field L-functions.

For the elliptic benchmark, the values of $x^3 - x$ at $x = 0, 1, 2, 3, 4$ modulo five are $0, 0, 1, 4, 0$. They give $1, 1, 2, 2, 1$ affine points and one point at infinity. Thus $\#X(\mathbb{F}_5) = 8$. The curve trace formula gives the degree-two numerator from trace -2 and determinant five. Its inverse roots are $-1 \pm 2i$, which verifies the required purity in this example directly. General purity is still the external theorem, not inferred from this finite check.

E Arithmetic realization and implementation

E.1 A finite endomorphism needs compatibility

For each i, choose vector-space complements $Z^i = B^i \oplus \widetilde{H}^i$ and $C^i = Z^i \oplus L^i$. The differential induces an isomorphism $L^i \to B^{i+1}$. A chain map has diagonal blocks for B^i, H^i, B^{i+1} in these adapted decompositions; it need not preserve the complements themselves. This proves the triangular factorization in Theorem 5.1 without assuming a Frobenius-equivariant splitting. It also shows why the endomorphism on an acyclic adjacent pair is conjugate through its differential: their characteristic determinants coincide. The Lean matrix lemma formalizes this conjugacy cancellation over an arbitrary commutative ring, rather than treating two independently chosen endomorphisms as equal.

The line functor and this characteristic function fit together only after the scalar realization is specified. For instance a perfect B-complex P with endomorphism F gives a determinant line construction over $B[u]$ and a rational trivialization from $1 - uF$ on its invertibility locus. The alternating characteristic expression can have poles: a one-dimensional term in cohomological degree one with $F = 1$ contributes $(1 - u)^{-1}$ to the determinant with exponent $(-1)^i$. It is not generally a global regular section under an arbitrary line convention. A basis change must transform both trivializations consistently. A noncommutative algebraic K-theory refinement is possible in other settings but would need its own definition and comparison; it is not the elementary determinant used in the Cartan formula.

For an infinite Hilbert-space operator $K(s)$, a Fredholm determinant requires trace class and holomorphic trace-class dependence. Continuity in trace norm can then justify convergence of

determinants. A zeta-regularized determinant of an unbounded operator requires its own spectral zeta continuation and normalization. These notions cannot be substituted without proof. An algebraic periodic resolution does not automatically define any of them.

E.2 A stronger interior duality counterexample

Even zeros strictly inside the strip can satisfy the expected reflection. Set $q = 16$, $F = \mathrm{diag}(2, 8)$ and the same alternating J. Then $F^t J F = 16J$ while

$$\det(1 - uF) = 1 - 10u + 16u^2 = (1 - 2u)(1 - 8u).$$

With $u = 16^{-s}$ its zeros have real parts $1/4$ and $3/4$. The determinant has the expected reciprocal pairing around $1/2$ yet fails purity. A positive Hermitian similitude cannot exist here, by Proposition 6.1. Consequently one cannot repair the arithmetic argument merely by enforcing the functional equation or Poincaré-type alternating duality.

E.3 A flow criterion and its missing construction

A conceivable global number-field object has a generator Θ with an inner product satisfying $\Theta + \Theta^* = 1$ on the zero-bearing cohomology, together with a rigorously defined regularized determinant for $s - \Theta$. If this identity is valid on an eigenvector v in the actual operator domains, then

$$(\rho + \overline{\rho})\langle v, v\rangle = \langle \Theta v, v\rangle + \langle v, \Theta v\rangle = \langle v, v\rangle,$$

so $\mathrm{Re}\,\rho = 1/2$. To turn this calculation into a zero theorem one must prove that determinant zeros correspond exactly to these eigenvalues, with no spurious factors or domain omissions. Deninger's foliated-space proposals [23] motivate this architecture but do not supply an established number-field realization to import. Cohomological special-value/determinant-line results such as Weil-étale constructions address different questions unless the missing all-zero identity and positivity are proved.

There are therefore two precise comparison targets: a determinant-preserving positive-pencil model of the existing convergent arithmetic approximants, or an independently built global arithmetic flow with the above spectral identity and regularization. Resolution maps can provide chain-level computational data and gluing obstructions for either target. Their existence alone proves neither comparison.

E.4 Implementation sequence with acceptance tests

The next work can proceed without repeatedly re-auditing the supplied papers.

1. Extend the actual Lean dual-number construction to idempotent path projectives and a categorical projective resolution for the terminating three-vertex example. Check graded shifts and Yoneda composition, not only scalar Betti recurrences. The required answer is the matrices and Ext product in Appendix A.

2. Build a marked-surface example with a declared dissection and line field, translate its string to actual maps, and compare its graded intersections with computed Hom groups including exceptional cases. Verify that the chosen realized object is perfect before any finite determinant.

3. Implement the conductor-patching functor in one split nodal example; check central and arrow-adic topologies, descent maps and derived specialization. Recover the extra one-dimensional H_1 in the nonflat fiber, rather than dropping it.

4. Construct a genuine arithmetic test category over $\mathbb{F}_5$ with an actual Frobenius realization and recover $1 + 2u + 5u^2$ from the elliptic benchmark. A curve's established cohomology is an external arithmetic dependency until formally constructed. An arbitrary fitted matrix with those coefficients would only reproduce a polynomial.

5. Before proposing a number-field theorem, write the exact functor, local trace and archimedean comparison. For primitive Dirichlet functions fix parity, primitive conductor and pole treatment at the principal case. For finite-order Hecke characters additionally fix field and infinity types. Dedekind and Artin cases require their own factorization and pole/holomorphy analysis; false integral induction from cyclic subgroups is not a replacement.

6. Attempt positivity or a determinant-preserving positive metric on the resulting object, with all-degree evidence and an independent adversarial proof review. A finite collection of positive minors, a finite zero check or an assumed metric positivity is not a universal theorem. If this step fails, record the exact obstruction and rotate to a different comparison construction.

7. Finally prove the analytic determinant convergence and exhaust the nontrivial strip, then apply Theorem 7.2. The published claim must track every remaining classical or informal dependency and the literal scope of Lean verification.

This sequence is complete as a specification of the missing constructions. Its arithmetic steps remain unresolved, so the present manuscript is a corrected foundation and conditional program rather than a completed GRH proof.

F Hodge cancellation and the boundary correction in detail

F.1 Harmonic representatives do not manufacture arithmetic weights

For a finite Hilbert complex,

$$\langle \Delta x, x\rangle = \|dx\|^2 + \|d^*x\|^2.$$

Thus $\ker \Delta = \ker d \cap \ker d^*$. In degree i, the finite-dimensional space of cycles decomposes orthogonally into B^i and $Z^i \cap (B^i)^\perp$. Since $(B^i)^\perp = \ker(d^{i-1})^*$, the second summand is precisely the harmonic space. It maps bijectively to cohomology. Alternating the dimensions cancels the boundaries, so $e - o = \chi(C) = \chi(H(C))$. This identifies the index power in Theorem 5.2; all paired nonzero eigenvalues have disappeared. Exactness is not needed for the normalized determinant identity.

For the fixed differential $d = 1$ in degrees zero and one, give the source Hermitian form $\overline{x}y$ and target form $r\overline{x}y$, $r > 0$. The adjoint identity forces $d^* = r$, so $\Delta_0 = \Delta_1 = r$. The ordinary factors $r - z$ move under $r = 1, 4$, whereas the normalized parity quotient remains one. This is a genuine change of positive metric on one fixed exact complex, not a change of its cohomology. It defeats an automatic identification of that ordinary determinant with an invariant arithmetic function.

The holomorphic issue is separate. For $d(s) = s$ and standard metrics, $\Delta(s) = |s|^2$. At $s = 1$, the difference quotient along real h is $2 + h$, while along ih, $h \in \mathbb{R}$, it is $-ih$. Their limits disagree. Consequently the Hodge operation does not preserve holomorphic dependence in a complex spectral parameter. A parameter-independent Hodge operator or a family self-adjoint only on a real slice has different hypotheses and still needs the exact arithmetic comparison.

An arithmetic endomorphism F commuting with d is not being discarded: its determinant on cohomology is the object in Theorem 5.1. The arithmetic weights of that F are independent information. Choosing harmonic representatives neither proves them nor supplies a positive form making F a scaled isometry.

Likewise analytic torsion uses degree weights: with a positive-Laplacian convention, one common expression is

$$T = \prod_i (\det{}' \Delta_i)^{(-1)^{i+1} i/2}.$$

It is not the unweighted parity superdeterminant. Ray–Singer [24], §1, Definition1.6 and Proposition1.7, introduce the regularized and finite combinatorial versions with their metric, acyclicity and basis hypotheses. Their Laplacian sign differs from ours. The finite cancellation theorem does not refute torsion, its geometric invariance theorems, or its possible arithmetic uses. Those would require a specified additional construction and comparison.

F.2 The relative operator is explicitly defined

On $H = \ell^2(\mathbb{N}) \otimes \mathbb{C}^r$, set $P = P_0 \otimes 1$ and $\widetilde{M} = 1 \otimes M$. The shift preserves the norm, its adjoint deletes the first coordinate, and $SS^* = 1 - P_0$. Put $A = 1 - \widetilde{M}$ and assume $1 - M$ invertible. Then

$$C = P + (1 - P)A, \qquad C^{-1} = P + (1 - P)A^{-1}, \qquad AC^{-1} = 1 - P\widetilde{M}.$$

The first two identities are verified on the orthogonal decomposition $PH \oplus (1 - P)H$. The last is multiplication on those same summands. Boundedness and invertibility follow from the finite matrix inverse; no unbounded-operator domain is suppressed. The perturbation has rank at most r. By the finite-support definition of its Fredholm determinant, its determinant is $\det(1 - M)$.

The Lean ring lemma proves the underlying algebraic product identity from explicit commutation and inverse hypotheses. It is valid in operator rings but does not itself construct the shift on mathlib's ℓ^2 space, prove trace class, or define the Fredholm determinant. Those are the prose constructions here. This distinction avoids claiming an infinite analytic formalization from a finite matrix theorem.

For the node, $R(-t)/R(t) = (1 + t)/(1 - t)$ reproduces its Cartan series. For an arbitrary locally gentle quiver, use $M = -V$ and $M = U$. The numerator continuation $\det(1 + V)$ is polynomial; division by $\det(1 - U)$ is meromorphic. Swapping the relative pair to $C_M A_M^{-1}$ reciprocates the determinant; merely reordering the multiplication does not. None of these operators is automatically positive or self-adjoint. In particular this realization leaves the positive-length obstruction of Proposition 7.1 intact.

F.3 The escaping terminal boundary prevents trace-norm convergence

Let S_N act on the first N coordinates and annihilate e_{N-1}. For $N \geq 2$,

$$S_N^* S_N = 1 - P_{N-1}, \qquad S_N S_N^* = 1 - P_0.$$

Using these in the same relative construction and extending by the identity on the infinite tail gives

$$R_N^{\mathrm{op}} = 1 - P_0 \otimes M + P_{N-1} \otimes \big((1 - M)^{-1} - 1\big).$$

The initial boundary acts by $1 - M$, the terminal one by $(1 - M)^{-1}$, and every other block is the identity. Therefore

$$\det R_N^{\mathrm{op}} = 1, \qquad \det_F(1 - P_0 \otimes M) = \det(1 - M).$$

For every $x \in H$, its coordinate norms tend to zero. Thus the last term tends strongly to zero. In trace norm, however,

$$\big\| R_N^{\mathrm{op}} - (1 - P_0 \otimes M) \big\|_1 = \big\| (1 - M)^{-1} - 1 \big\|_1.$$

Embedding a finite matrix on a coordinate subspace preserves its nonzero singular values, proving the equality. The right side is positive for $M \neq 0$: if it vanished, the inverse would be the identity and hence $M = 0$. For $M = 1/2$ it is exactly one. This is the precise failed determinant-continuity step, not a contradiction with Sylvester's finite identity.

There is also no unrestricted infinite heat-cancellation theorem from a trace-class *difference* alone. For the unilateral shift,

$$e^{-tS^*S} - e^{-tSS^*} = (e^{-t} - 1)P_0,$$

whose trace is $e^{-t}-1$, while index $S = -1$. Neither individual heat operator is trace class. Compatible trace-class heat or specified relative regularization hypotheses must be verified; one cannot cancel two divergent traces. For Hilbert–Schmidt B, by contrast, both Gram perturbations of the identity are trace class and the corresponding Fredholm Sylvester identity has its usual well-defined setting. This does not cover the bulk shift construction.

The next arithmetic comparison must specify its terminal/boundary correction and prove convergence in the topology required by its determinant. Removing the terminal term makes the relative determinant $\det(1 - M)$ at each stage, but changes the finite realization and requires an independent arithmetic interpretation. We have constructed an exact relative Cartan model and isolated a failed natural approximation. The number-field determinant, positive metric for that same function and the resulting GRH proof remain open.

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